

Weekly Report

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Motivation: Why Algebraic K-Theory for ER?

- Algebraic K-theory studies **equivalence**, **exactness**, and **homotopy consistency**.
- Entity Resolution (ER) also tries to construct **equivalence classes of real-world entities**.
- Local comparisons (record pairs) often contradict each other.
- K-theory provides a natural mathematical language for:
 - resolving contradictions,
 - building global invariants,
 - ensuring structural consistency.

Core Correspondence: K-Theory $\leftrightarrow$ ER

K-theory concept	Implication in ER
Exact sequence	Compatibility of different fields
Homotopy equivalence	Different match paths should agree
Higher K-groups	Global invariants of clusters
Localization	Ignore noisy/irrelevant attributes
Pushout / Pullback	Merge or split clusters formally

Thus, ER can be viewed as constructing a **homotopy-consistent equivalence structure**.

Commutativity as Consistency

Given three records A, B, C and morphisms:

$$A \xrightarrow{f_{\text{title}}} B \xrightarrow{g_{\text{auth}}} C \quad \text{vs.} \quad A \xrightarrow{h_{\text{year}}} C$$

- Traditional ER only sees conflicting similarity scores.
- K-theory/CEER checks whether the diagram commutes:

$$g_{\text{auth}} \circ f_{\text{title}} \simeq h_{\text{year}}$$

- If not: **homotopy inconsistency** $\Rightarrow$ cannot be in one cluster.

Example 1: Cora Dataset (Non-Commuting Triangle)

Scenario

- A, B similar by *title*.
- B, C similar by *authors*.
- A, C conflict by *year*.

Interpretation

- Traditional ER: three inconsistent pairwise similarities.
- CEER/K-theory:

$$g \circ f \not\approx h$$

The triangle does not commute.

- $\Rightarrow$ structural reason to reject the match.

Example 2: Affiliation Dataset (Pushout = Cluster Merge)

Records:

- A: “MIT CSAIL”
- B: “Massachusetts Institute of Technology”
- C: “CSAIL, MIT, Cambridge”

K-theory intuition:

- If $A \simeq B$ and $A \simeq C$, then B and C must merge via a **pushout**.
- CEER gives a formal cluster merge justified by morphisms, not heuristics.

Example 3: D10K Synthetic Dataset (Localization Removes Noise)

Noise challenge:

- Random perturbations in phone, zipcode, street numbers.
- Many weak edges in the similarity graph.

Localization (K-theory):

- Treat low-confidence morphisms as “non-invertible” and remove them.
- Stabilizes homotopy type of similarity graph.
- Produces more reliable cluster assignments.

Example 4: North Carolina Voter Registry (Exactness Test)

Common scenario:

- Same name, inconsistent birthdate.
- Address changes across counties.

Exactness interpretation:

- If $A \rightarrow B$ (name supports match)
- but $A \rightarrow C$ (birthdate contradicts)
- then the diagram breaks exactness.

Result: A cannot simultaneously align with B and C.

Big Picture: Why This Matters for ER Research

- Provides a **mathematically principled** framework for ER.
- Naturally integrates with Categorical Embedding for Entity Resolution (CEER), categorical Potts models (CPM), and functorial embeddings.
- Allows ER to:
 - resolve contradictions structurally,
 - enforce global consistency,
 - justify merges/splits with algebraic meaning.
- Leads to a new methodology:

Homotopy-Consistent Entity Resolution

- Develop categorical embeddings trained on Cora/Affiliation ground truth.
- Implement homotopy-consistency loss.
- Extend CEER to pushout/pullback-based clustering.
- Aim at publications in KDD, VLDB, TKDE.

Thank You!

To be continued...